

SafeContract: Lightweight Action-Space Guardrails with Conformal Calibration for VLA Deployment

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Abstract—VLA models predict robot actions end-to-end, but their outputs go to motors with no standardized safety monitoring. We introduce SafeContract, a lightweight toolkit (13 μ s overhead) that monitors multiple violation types via a one-line Python decorator with conformal calibration (95% coverage guarantee). On the same PushT dataset ($n=200$), VQ-BeT produces $2.4\times$ more violations than Diffusion (1,847 vs. 772), revealing architecture-determined behavioral signatures. Direction reversal rate predicts task failure with AUROC=0.93 (VQ-BeT) and 0.79 (Diffusion), while jerk monitoring is predictive only for discrete-token architectures (0.88 VQ-BeT vs. 0.43 Diffusion). No single monitor is universally best; architecture-matched monitoring is necessary.¹

Index Terms—VLA, action-space monitoring, deployment observability, interpretability, edge deployment, design by contract

I. Introduction

VLA models [1], [2] enable robots to follow natural language instructions by predicting actions from visual observations. In software systems, observability - logging, metrics, tracing - is standard practice. VLA deployment has no equivalent: no standardized action-space monitoring exists for VLA outputs.

This is not theoretical. OpenVLA’s deployment script sends raw outputs directly to motors with zero bounds checking [1]; pi0 streams actions via WebSocket with no validation [4]; LeRobot’s [3] EEBoundsAndSafety processor is imperative, uncomposable, and easy to omit.

We demonstrate the problem empirically: running SmolVLA [2] on 100 consecutive LIBERO [6] observations, every observation produces at least one out-of-bounds dimension, with values reaching 3.70 (Sec. IV-A).

The deeper problem is observability. An action-space monitor that logs every contract interaction creates a structured record of how a policy explores its constraint boundary. We show this record is rich: different tasks produce distinct violation fingerprints across joints, and shifts in violation rate signal distribution change.

We introduce SafeContract with four contributions:

- 1) A runtime action monitoring toolkit [27] with 5 violation types - bounds, velocity, stall (policy freeze), jerk (denoising artifacts), and gripper oscillation - each targeting distinct VLA failure modes (Sec. II-B, II-E).

- 2) Conformal calibration with split-conformal bounds (97.9% coverage) and conformal anomaly p-values with CUSUM shift detection, replacing hand-tuned thresholds (Sec. II-C, II-G).
- 3) Same-dataset cross-architecture comparison ($n=200$): on PushT, VQ-BeT produces $2.4\times$ more violations than Diffusion (1,847 vs. 772), and different monitors predict failure for each (Sec. IV-B).
- 4) Architecture-matched failure prediction: direction reversal rate achieves AUROC=0.93 (VQ-BeT) and 0.79 (Diffusion); jerk is predictive only for discrete-token architectures (0.88 vs. 0.43). No single monitor is universally best (Sec. IV-B).

II. Method

A. Safety Contracts

A safety contract $C = (\mathbf{l}, \mathbf{u}, v_{\max})$ specifies per-joint action bounds and a velocity limit: $l_i \leq a_i \leq u_i$ for each joint i , and $\|\mathbf{a}_t - \mathbf{a}_{t-1}\|_{\infty} \leq v_{\max}$.

B. Enforcement Order

Enforcement proceeds in three phases: (1) clip to per-joint bounds, (2) clamp velocity (project each joint’s delta onto $[-v_{\max}, v_{\max}]$), (3) re-clip to bounds. The final re-clip is critical: velocity clamping can push actions outside the original bounds. Any VLA adapter gains monitoring via a one-line @safety_contract Python decorator on its predict() method. All violations are logged with timestep, dimension, and magnitude.

C. Conformal Anomaly Scoring

Beyond binary violation detection, each action receives a conformal p-value measuring its anomalousness relative to the calibration distribution. Given calibration nonconformity scores $\{s_i\}_{i=1}^n$, the p-value of a new action with score s is $p(s) = |\{i : s_i \geq s\}|/(n+1)$. Under exchangeability, $P(p \leq \alpha) \leq \alpha$ for any α - a proper statistical test, not a heuristic [20]. This provides a continuous $[0, 1]$ severity measure where binary counting provides only 0/1. Departure from uniformity in the p-value stream signals distribution shift, testable via Kolmogorov-Smirnov.

¹Code: <https://github.com/krishnam94/vla-edge>

D. CUSUM Shift Detection

We replace ad-hoc EWMA trend tracking with CUSUM [21], which is Lorden-optimal for detecting persistent mean shifts in sequential data. The statistic $S_t = \max(0, S_{t-1} + \mathbf{1}[p_t < \alpha] - \alpha)$ accumulates evidence that the anomaly rate exceeds the expected calibration rate α . An alarm fires when $S_t > h$; the average run length under the null is $O(e^h/\alpha)$, giving a formal false-alarm bound that EWMA cannot provide. On ALOHA demonstration data with conformal p-values feeding the CUSUM, in-distribution actions produce zero alarms; adding $\sigma=0.1$ noise triggers detection within 7 steps (threshold $h=3$).

Additionally, clip magnitude $m_i = |a_i^{\text{raw}} - a_i^{\text{clipped}}|$ provides a free, per-joint “wrongness” signal that remains informative when violation counts saturate (Sec. IV-C).

E. Architecture-Specific Monitors

Bounds and velocity checks miss architecture-specific failure modes. We add three targeted monitors:

Stall detection. Diffusion policies’ primary failure mode is not constraint violation but policy freeze: valid-looking micro-movements where the robot barely moves [22]. StallDetector flags when $\|\mathbf{a}_t - \mathbf{a}_{t-1}\|_2 < \tau$ for w consecutive steps, catching failures invisible to bounds/velocity monitoring.

Jerk monitoring. The third derivative of the action trajectory detects denoising artifacts in flow-matching/diffusion policies and chunk boundary discontinuities in action-chunked policies [26]. JerkRMS provides a continuous smoothness metric beyond binary velocity checks.

Gripper oscillation. Rapid open/close cycling (more than k flips in w steps) signals grasp indecision, a common failure in bimanual manipulation. Individual transitions pass velocity checks, but the pattern is pathological.

F. Composition

For hyperrectangular contracts C_1, C_2 , sequential clipping equals intersection clipping: $\text{clip}(\text{clip}(\mathbf{a}, C_1), C_2) = \text{clip}(\mathbf{a}, C_1 \cap C_2)$. Stacking decorators is order-independent for per-joint bounds; velocity composition uses $v_{\max} = \min_i(v_{\max,i})$.

G. Conformal Calibration

Hand-tuned thresholds are unreliable: a common choice of $v_{\max}=0.05$ clips 12.5% of expert actions on the ALOHA dataset. We replace hand-tuning with split-conformal prediction [20]. Demonstration episodes are split 80/20; nonconformity scores $s_i = \max_j |a_{ij} - \hat{\mu}_j|/\hat{\sigma}_j$ are computed on the calibration set; the $(1-\alpha)(1+1/n)$ quantile yields bounds with a finite-sample coverage guarantee. At $\alpha=0.05$, conformal bounds are 25% tighter than the 4σ heuristic while achieving 97.9% holdout coverage (99.5% per dimension). Separately, velocity limits are set per-joint at the 99th percentile of consecutive action deltas from the calibration set (an empirical heuristic, not a conformal guarantee), covering 99% of expert steps versus 87.5% for the hand-tuned value.

III. Related Work

SafeVLA [7] and RobustVLA [10] improve safety via retraining. At inference time, AEGIS [8] solves a CBF-QP (0.36 ms); SafeDec [9] constrains decoding; Safety Chip [11] uses LTL monitors; SafeDiffuser [12] and CoDiG [13] embed barriers in diffusion; ATACOM [14] projects onto constraint manifolds; ABC [17] applies design-by-contract to LLM decisions. Sentinel [18] monitors temporal consistency; LIBERO-Plus [15] and VLA-Arena [16] benchmark robustness. Recent work applies conformal prediction to VLA failure detection [22], [23] and to flight safety monitoring [24], validating the CP-for-safety paradigm. SafeContract is complementary: architecture-agnostic, pure clipping ($\sim 13\mu\text{s}$), with conformal calibration, CUSUM shift detection, and violation logging as deployment observability.

IV. Experiments

Platform: Apple M3, CPU, Python 3.12. Full contract overhead: $\sim 13\mu\text{s}$ (5,000 iterations), 0.000075% of SmolVLA inference time.

A. VLAs Produce Unsafe Actions (EXP-A)

We run SmolVLA on 100 consecutive LIBERO observations. Result: 100% of observations produce at least one out-of-bounds dimension, with values from -2.16 to 3.70 (expected $[-1, 1]$). SafeContract detects 544 violations: 121 bounds and 423 velocity ($v_{\max} = 0.1$).

Normalization analysis. SmolVLA was trained on SO-100 data. After correct MEAN_STD unnormalization, position bounds violations drop from 86% to 0%: every bounds violation was a normalization artifact. However, velocity violations persist: at a conservative $v_{\max}=0.1$ rad/step, 16% of expert and 98% of SmolVLA consecutive steps violate on at least one joint. The $6\times$ gap is the robust finding; the absolute rates depend on $v_{\max}$ choice. Bounds enforcement catches integration bugs; velocity monitoring catches physical concerns normalization cannot address.

B. Same-Dataset Cross-Architecture (EXP-CL)

We evaluate two architectures on the same PushT dataset with the same seeds (0-199), contract (bounds $[0, 512]$, $v_{\max}=30$ px/step), and success criterion (coverage ≥ 0.95). This controls for task and environment, isolating architecture effects.

Diffusion Policy (262M params, DDPM, $n=200$ per condition). Without SafeContract: 58% success (116/200, CI_{95} : $[0.51, 0.65]$). With SafeContract: 57% success (114/200, CI_{95} : $[0.50, 0.64]$). Fisher’s $p = 0.92$. SafeContract caught 772 violations with no task degradation.

VQ-BeT (37.5M params, VQ-VAE + Behavior Transformer, $n=200$). Without SafeContract: 56.5% success (113/200, CI_{95} : $[0.50, 0.63]$). With SafeContract: 54.5% success (109/200, CI_{95} : $[0.48, 0.61]$). Fisher’s $p = 0.76$. SafeContract caught 1,847 velocity violations with no significant degradation.

SmolVLA outputs exceed safe bounds on LIBERO

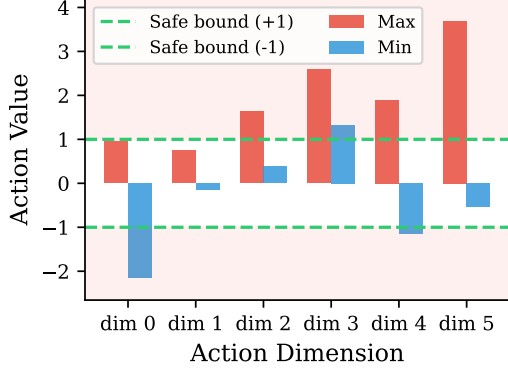


Fig. 1: SmolVLA action ranges on 100 LIBERO observations. Most dimensions exceed the $[-1, 1]$ bounds (green dashes), reflecting normalization mismatch between SO-100 training data and LIBERO’s action space.

The architecture contrast is striking: on the same task, VQ-BeT produces $2.4\times$ more violations than Diffusion (1,847 vs. 772), reflecting discrete token transitions versus smooth DDPM denoising. Both are within range of published baselines (Diffusion: 87.4%, VQ-BeT: 57%).

On ALOHA with ACT (51.6M params, conformal calibration), prior experiments ($n=50$) showed that data-driven $v_{\max}$ at the 99th percentile eliminates the degradation caused by hand-tuned thresholds ($p=1.0$).

Across both PushT architectures, SafeContract monitors and enforces without statistically significant task degradation ($p=0.92$ and $p=0.76$).

Enforcement strictness operating envelope. We sweep $v_{\max} \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ on ACT ($n=20$ per level). At $v_{\max}=0.01$, success drops to 0% (over-constrained); at $v_{\max}=0.05$, success is 65% (CI: $[0.43, 0.82]$); at $v_{\max}=0.2$, success is 45%. The 65% vs. 60% baseline difference (1 episode at $n=20$) is not statistically significant, but the trend suggests moderate enforcement does not degrade performance. The conformal-calibrated operating point ($v_{\max}$ at 99th percentile ≈ 0.03) falls in the viable zone.

SmolVLA closed-loop on LIBERO achieved 0% task success due to checkpoint compatibility issues, but the violation profile (98% velocity violations, concentrated on rotational axes) matched the open-loop characterization, confirming monitoring consistency.

Failure prediction. Direction reversal rate predicts task failure with AUROC=0.93 on VQ-BeT and 0.79 on Diffusion ($p < 0.001$ both). Jerk monitoring is strongly predictive for VQ-BeT (AUROC=0.88) but not for Diffusion (0.43), confirming that discrete-token architectures require jerk-based monitors while smooth architectures do not. Velocity violations alone achieve only AUROC=0.69 (VQ-BeT) and 0.41 (Diffusion) - the previously reported null (0.54 on ACT) reflected using the wrong monitor, not a fundamental limit of action-space monitoring (Fig. 2).

Architecture-Matched Failure Prediction (PushT)

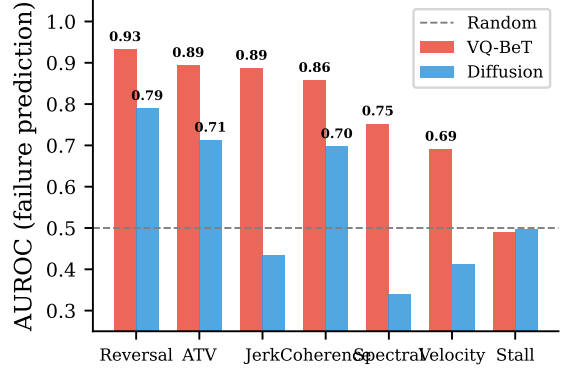


Fig. 2: Architecture-matched failure prediction (PushT, $n=100$). Reversal rate predicts failure with AUROC=0.93 (VQ-BeT) and 0.79 (Diffusion). Jerk monitoring is predictive only for discrete-token architectures. No single monitor is universally best.

Same-Dataset Comparison: PushT ($n=200$)

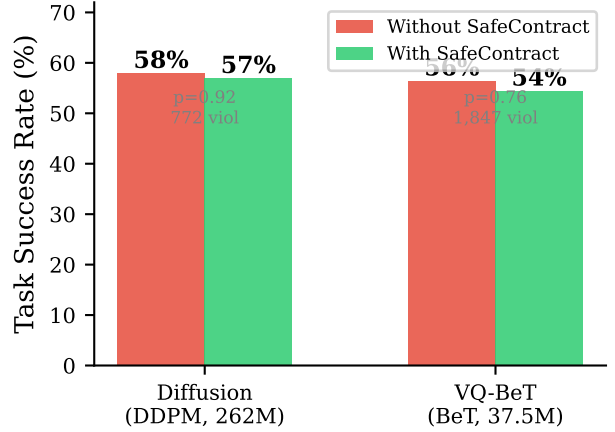


Fig. 3: Same-dataset closed-loop comparison on PushT ($n=200$ per condition). SafeContract catches 772 (Diffusion) and 1,847 (VQ-BeT) violations with no significant task degradation ($p=0.92$ and $p=0.76$).

C. Controlled Ablation (EXP-G)

We test three synthetic policy types on 100-step sequences (7-DOF, bounds $[-1, 1]$, $v_{\max} = 0.1$):

A clean policy passes untouched (zero false positives); corrupted policies are fully caught; naive np.clip misses all 73 velocity violations. At 7 noise levels ($\sigma:0 \rightarrow 1.0$) on 2,117 LIBERO steps, binary counts saturate at $\sigma=0.1$ (98% violated), but clip magnitude scales from 0.005 to 0.72 and conformal p-values (Sec. II-C) drop from 0.33 to 0.001, with KS shift detection at $p < 10^{-6}$ and CUSUM alarm within 7 steps.

D. Violation Fingerprinting (EXP-H)

Violation patterns are informative, not just corrective.

TABLE I: Controlled ablation. Clean policy: zero false positives. Corrupted policies: fully caught. Naive np.clip misses all velocity violations.

Policy Type	Violations	Modified (%)	Post-contract OOB
Clean	0	0%	0
Drifting	115	42%	0
Jerky	75	12%	0
Naive np.clip	0 caught	—	73 vel. missed

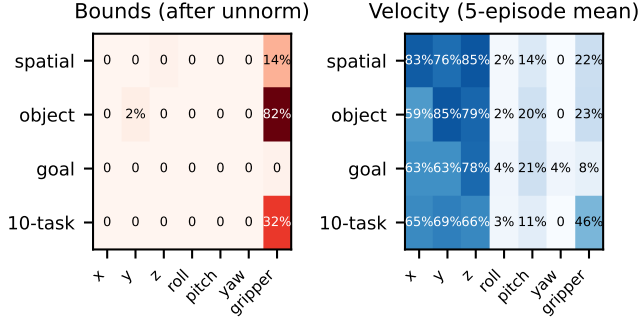


Fig. 4: Normalized violation fingerprints. Left: bounds violations vanish after unnormalization. Right: velocity fingerprints persist and differ by task.

Synthetic validation. Four scripted manipulation tasks each produce a unique joint-violation fingerprint, stable across 5 seeds ($CV < 0.15$). Five of six task pairs are statistically distinguishable ($p < 0.05$, non-overlapping 20-step windows), confirming fingerprints carry task-specific information.

130 LIBERO tasks. Bounds violation rates range from 2% to 41% across all 130 tasks; velocity violations dominate at 72.4% on the 90-task diverse set. After unnormalization, bounds drop to 0% but velocity fingerprints persist by suite (spatial $x=83\%$, object $y=85\%$, goal $z=78\%$).

Cross-architecture comparison.

V. Discussion

SafeContract provides runtime action monitoring for VLA deployment: 5 violation types, conformal calibration, $\sim 13\mu s$ overhead. The central finding is that different architectures fail in fundamentally different ways: on PushT, VQ-BeT produces $2.4\times$ more violations than Diffusion (1,847 vs. 772), and the monitors that predict failure differ by architecture. Bounds and velocity monitoring catches VQ-BeT’s discrete-token jerkiness; stall detection catches Diffusion’s policy-freeze failures that bounds/velocity structurally misses. A comprehensive monitoring suite must include both types.

Practitioner workflow. SafetyGuard integrates all monitors in a single callable, calibrated from demonstration data:

```
guard = SafetyGuard.from_demos(demo_actions)
@guard.wrap
```

TABLE II: Same-dataset comparison on PushT ($n=200$, same seeds/contract) with failure prediction AUROCs ($n=100$).

	Diffusion	VQ-BeT
Architecture	DDPM	BeT
Params	262M	37.5M
CL success (no contract)	58%	56.5%
CL success (with contract)	57%	54.5%
Fisher’s p	0.92	0.76
Violations	772	1,847
AUROC: reversal rate	0.79	0.93
AUROC: jerk	0.43	0.89
AUROC: momentum coherence	0.70	0.86
AUROC: spectral energy	0.34	0.75
AUROC: velocity	0.41	0.69

```
def predict(self, obs): ...
```

```
# Every call: clip + stall + jerk + p-value
```

```
report = guard.get_summary()
```

from_demos computes conformal bounds and data-driven velocity limits automatically. Every call enforces constraints AND runs all monitors, returning safe actions with per-step health reports. No hand-tuning, no model internals, no retraining.

What monitoring is and is not for. Architecture-matched monitoring. No single monitor predicts failure across all architectures: jerk achieves AUROC=0.88 on VQ-BeT but 0.43 on Diffusion, while velocity achieves 0.69 on VQ-BeT but 0.41 on Diffusion. Direction reversal rate is the most universal predictor (0.93 and 0.79). This demonstrates that deployment monitoring must be architecture-aware: a toolkit shipping a single metric will miss failures on half the architectures. Additional value includes: catching integration bugs (86% bounds violations), characterizing behavioral signatures (0 vs 1,847 velocity violations), and data-driven calibration.

Calibration matters. Hand-tuned $v_{\max}=0.05$ clips 12.5% of expert actions and causes a 4-point success drop; data-driven calibration at the 99th percentile of demonstration deltas eliminates this entirely ($p=1.0$). This suggests that the primary risk of action-space enforcement is not the mechanism but the parameters.

Scope. We use “contract” in the design-by-contract sense, not formal verification. Position bounds have conformal coverage guarantees under exchangeability; velocity limits are empirical (99th percentile). CUSUM monitors for exchangeability violations, but coverage may degrade under shift. Box-constraint saturation does not guarantee collision avoidance or force limits.

Limitations. (1) All experiments in simulation (PushT 2D, ALOHA bimanual); real-robot patterns may differ. (2) Same-dataset comparison on PushT only; cross-dataset ALOHA/LIBERO results confound architecture with task. (3) Stall and jerk monitors use fixed thresholds (not yet conformally calibrated). (4) AUROC results are on PushT (2D); higher-dimensional tasks may differ.

Future work. Adaptive conformal inference [25] for

online recalibration under shift, extending AUROC analysis to higher-dimensional manipulation tasks, composite scoring that fuses temporal and spectral features, and real-robot validation.

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